

Experiment - 9

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Title: Design input forms with validation using
HTML, CSS, JavaScript

AIM: To design an input form that collects email and phone number.

Step 1: HTML Form Creation

Code: HTML (index.html)

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>Experiment 9: Form Validation</title>
```

```
<link rel="stylesheet" href="style.css">
```

```
</head>
```

```
<body>
```

```
<div class="container">
```

```
<h2>Registration Form</h2>
```

```
<form id="myForm">
```

```
<div class="form-group">
```

```
<label for="email">Email:</label>
```

```
<input type="text" id="email" placeholder="Enter email">
```

```
<small id="emailError" class="error-message"></small>
```

```
</div>
```

```
<div class="form-group">
```

```
<label for="phone">Phone Number:</label>
```

```
<input type="text" id="phone" placeholder="Enter phone number">
```

```
<small id="phoneError" class="error-message"></small>
```

```
</div>
```

```
<button type="submit">Submit</button>
```

```
</form>
```

```
</div>
```

```
<script src="https://cdn.jsdelivr.net/npm/validator@13.11.0/validator.min.js"></script>
```

```
<script src="script.js"></script>
```

</body>

</html>

OUTPUT:

Registration Form

Email:

Phone Number:

Explanation:

- Created a structured registration form using HTML.
- Includes email and phone number input fields
- `<small>` tags used to display error messages
- Linked external CSS and JavaScript files
- No styling or validation applied yet

Step 2: CSS Styling

Code: CSS (style.css)

```
/* Global styles for the entire page */
* {
  margin: 0; padding: 0;
  box-sizing: border-box;
}

/* Body styling - center content and add background color */ body
{
  font-family: 'Arial', sans-serif;
  background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
  display: flex; justify-
content: center; align-
items: center; min-
height: 100vh; padding:
20px;
}

/* Container styling - card layout with rounded corners and shadow */
.container { background-
color: white; padding: 40px;
border-radius: 15px; box-
shadow: 0 10px 25px rgba(0,
```

```
0, 0, 0.2); width: 100%;  
max-width: 400px;  
}
```

```
/* Heading styling */ h2  
{ text-align: center;  
color: #333; margin-  
bottom: 30px; font-  
size: 28px;  
}
```

```
/* Form group styling for each input field */  
.form-group { margin-  
bottom: 25px;  
}
```

```
/* Label styling */  
label { display:  
block; margin-  
bottom: 8px; color:  
#555; font-weight:  
600; font-size: 14px;  
}
```

```
/* Input field styling */ input {  
width: 100%; padding: 12px;  
border: 2px solid #ddd; border-
```

```
radius: 8px; font-size: 16px;  
transition: border-color 0.3s ease;  
}
```

```
/* Input focus effect */  
input:focus { outline:  
none; border-color:  
#667eea;  
}
```

```
/* Valid input - green border (bonus feature) */  
input.valid { border-color: #28a745;  
}
```

```
/* Invalid input - red border (bonus feature) */  
input.invalid { border-color: #dc3545;  
}
```

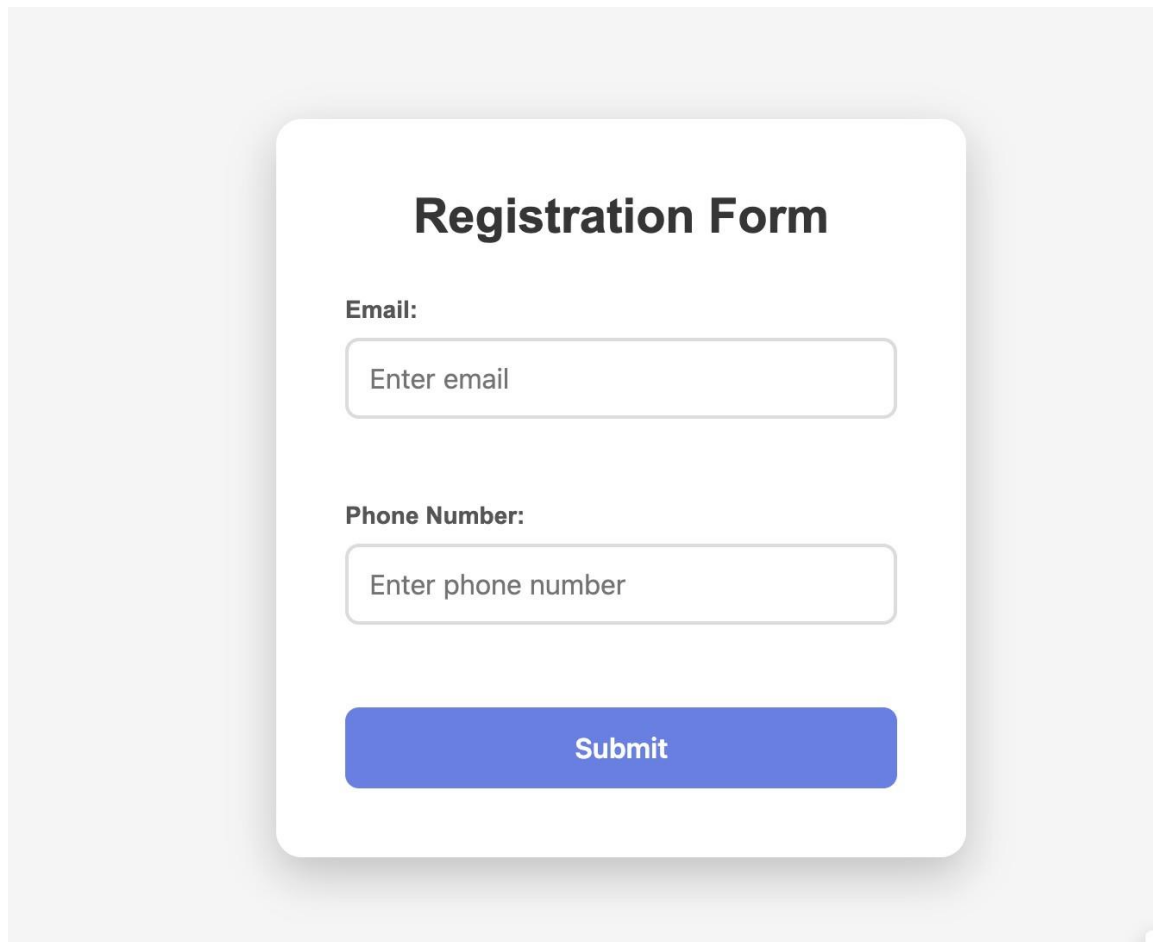
```
/* Error message styling - red color */ .error-message {  
display: block;  
color: #dc3545;  
font-size: 13px;  
margin-top: 5px;  
min-height: 18px;  
}
```

```
/* Submit button styling */ button { width: 100%; padding: 14px;
background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
color: white; border: none; border-radius: 8px;
font-size: 16px; font-weight: 600; cursor: pointer;
transition: transform 0.2s ease, box-shadow 0.2s ease;
}
```

```
/* Button hover effect */ button:hover { transform:
translateY(-2px); box-shadow: 0 5px 15px
rgba(102, 126, 234, 0.4);
}
```

```
/* Button active effect */
button:active { transform:
translateY(0);
}
```

OUTPUT:



The image shows a registration form centered on a light gray background. The form is a white card with rounded corners and a subtle shadow. At the top of the card is the title "Registration Form" in a bold, dark font. Below the title are two input fields. The first is labeled "Email:" and contains the placeholder text "Enter email". The second is labeled "Phone Number:" and contains the placeholder text "Enter phone number". Both input fields have a light blue border and rounded corners. At the bottom of the card is a blue button with rounded corners and the text "Submit" in white.

Explanation:

- Applied modern UI styling using CSS
- Form is centered with gradient background
- Card layout with shadow and rounded corners
- Input fields show color feedback (valid/invalid)
- Error messages displayed in red
- Improves overall user experience

Step 3: JavaScript Validation

Code: JavaScript (script.js)

```
document.addEventListener('DOMContentLoaded', function() {
```

```
    const form = document.getElementById('myForm'); const
```

```
    emailInput = document.getElementById('email'); const
```

```
    phoneInput = document.getElementById('phone'); const
```

```
    emailError = document.getElementById('emailError'); const
```

```
    phoneError = document.getElementById('phoneError');
```

```
    function validateEmail(email) { return
```

```
        validator.isEmail(email);
```

```
    }
```

```
    function validatePhone(phone) { return
```

```
        validator.isMobilePhone(phone, 'en-IN');
```

```
    }
```

```
    function showError(inputElement, errorElement, message) {
```

```
        errorElement.textContent = message;
```

```
        inputElement.classList.add('invalid');
```

```
        inputElement.classList.remove('valid');
```

```
    }
```

```
function showValid(inputElement, errorElement) {  
  errorElement.textContent = ''; inputElement.classList.remove('invalid');  
  inputElement.classList.add('valid');  
}
```

```
function resetInput(inputElement, errorElement) {  
  errorElement.textContent = ''; inputElement.classList.remove('invalid');  
  inputElement.classList.remove('valid');  
}
```

```
emailInput.addEventListener('input', function() { const  
  email = emailInput.value.trim();
```

```
  if (email !== '') { if  
    (validateEmail(email)) {  
      showValid(emailInput, emailError);  
    } else { showError(emailInput, emailError, 'Please enter a valid email  
      address');  
    }  
  } else { resetInput(emailInput,  
    emailError);  
  }  
});
```

```
phoneInput.addEventListener('input', function() { const  
  phone = phoneInput.value.trim();
```

```
if (phone !== "") { if
(validatePhone(phone)) {
showValid(phoneInput, phoneError);
} else {
showError(phoneInput, phoneError, 'Please enter a valid Indian phone number');
}
} else {
resetInput(phoneInput, phoneError);
}
});
```

```
form.addEventListener('submit', function(event) {
event.preventDefault();
```

```
const email = emailInput.value.trim(); const
phone = phoneInput.value.trim();
```

```
let isValid = true;
```

```
if (email === "") { showError(emailInput, emailError,
'Email is required'); isValid = false;
} else if (!validateEmail(email)) {
showError(emailInput, emailError, 'Please enter a valid email address');
isValid = false;
} else {
```

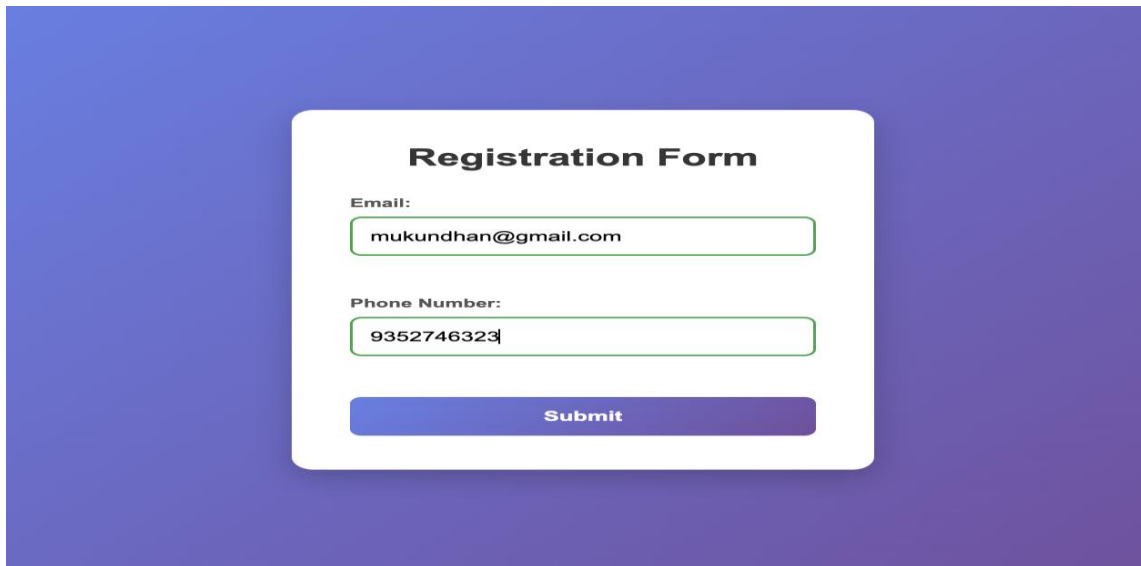
```
showValid(emailInput, emailError);  
}
```

```
if (phone === "") { showError(phoneInput, phoneError, 'Phone  
number is required');  
isValid = false;  
} else if (!validatePhone(phone)) { showError(phoneInput, phoneError, 'Please  
enter a valid Indian phone number'); isValid = false;  
} else {  
showValid(phoneInput, phoneError);  
}
```

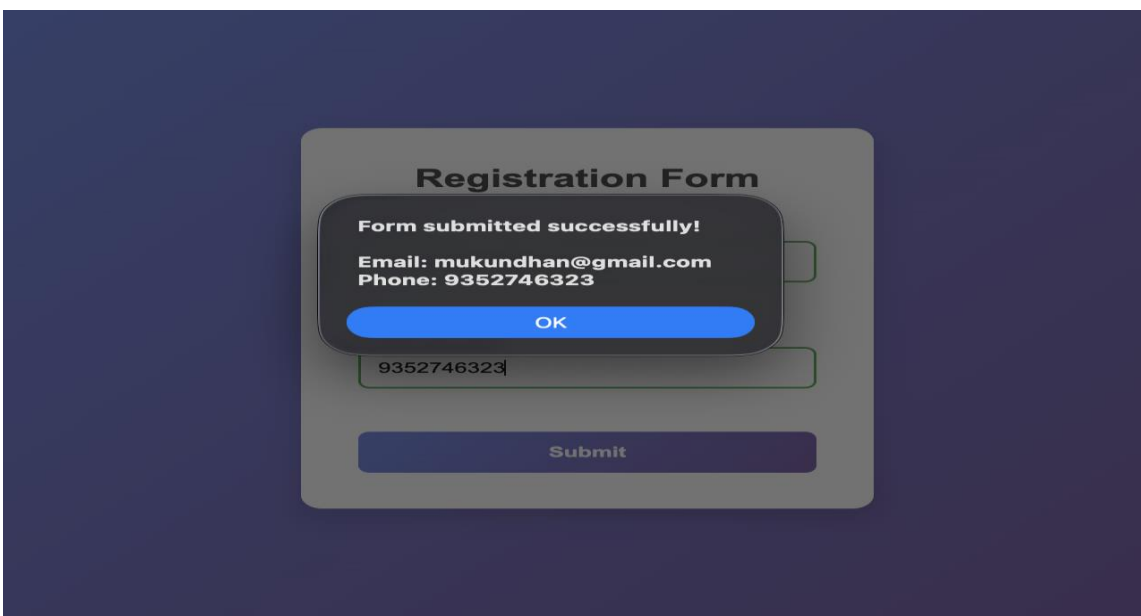
```
if (isValid) { alert('Form submitted successfully!\n\nEmail: ' + email +  
\nPhone: ' + phone);
```

```
form.reset(); resetInput(emailInput,  
emailError); resetInput(phoneInput,  
phoneError);  
}  
});  
});
```

OUTPUT:



A registration form titled "Registration Form" is displayed on a purple gradient background. The form is white with rounded corners and a shadow. It contains two input fields: "Email:" with the value "mukundhan@gmail.com" and "Phone Number:" with the value "9352746323". Below the inputs is a blue "Submit" button.



The same registration form is shown, but with a dark gray modal overlay. The modal has a title "Registration Form" and a message "Form submitted successfully!". It displays the submitted email "Email: mukundhan@gmail.com" and phone "Phone: 9352746323". Below the message is a blue "OK" button. The form's "Submit" button is now disabled and grayed out.

Explanation:

- Used **Validator.js** for email and phone validation
- Implemented real-time validation (bonus feature)
- Displays dynamic error messages
- Prevents form submission if inputs are invalid

- Shows success alert on valid input
- Enhances data accuracy and user interaction